Vansh garg

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B.Tech (ECE)
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Education

Indian Institute of Information Technology Ranchi CGPA: 7.96/10.0
B.Tech in Electronics and Communication Engineering

Personal Projects

HFSS Microstrip Antenna Design Automation using PyAEDT May 2026
Tools: Ansys HFSS, PyAEDT, Python, NumPy, Matplotlib GitHub

- Engineered a fully automated antenna design pipeline using PyAEDT to eliminate manual HFSS GUI workflows
- Reduced antenna prototyping time from hours to under 1 minute via a 6-parameter array-driven architecture
- Implemented parametric geometry generation (ground, substrate, patch, feed, slot) with dynamic frequency sweep
- Automated post-processing pipeline extracting S11, VSWR, Smith Chart, radiation patterns, and bandwidth analysis
- Designed and optimized a 2.4 GHz microstrip antenna achieving S11 < -10 dB and VSWR < 1.5 using automated HFSS workflows
- Delivered real-time field visualization (E-field, H-field, current distribution) and automated report generation

Custom IR Sensor PCB Design March 2025
Tools: KiCad, LTspice, JLCPCB, Soldering

- End-to-end PCB design in KiCad: schematic capture, component placement, and 2-layer routing
- Integrated IR transmitter-receiver pair with comparator circuitry for digital/analog dual output
- Optimized trace widths and ground plane layout to reduce noise and ensure signal integrity
- Validated design via DRC/ERC; generated fabrication files and successfully assembled prototype
- Tested sensor response — stable detection range of 10–20cm with adjustable sensitivity control

ESP32-Based Bluetooth Radar System Jan 2025
Tools: ESP32, Arduino IDE, C++, HC-SR04, Servo, Bluetooth Serial, TM1637 GitHub

- Developed a 180° scanning radar system on ESP32 for real-time obstacle detection and ranging
- Engineered servo-sensor synchronization logic to map distance data across a polar coordinate grid
- Integrated TM1637 display and piezo buzzer for dual-mode feedback — visual distance + audible alerts
- Enabled wireless control via Bluetooth Serial; designed custom Android terminal for remote operation
- Tuned ultrasonic pulse timing and servo delay parameters to eliminate echo interference and false readings
- Packaged entire system into a compact 3D-printed enclosure with clean cable management

Technical Skills

Programming Languages: Python, Embedded C, C++, HTML, CSS

Microcontroller Boards: Arduino Uno, Arduino Nano, Arduino Mega, ESP32, ESP32-S3, NodeMCU (ESP8266), STM32

Communication Protocols: UART, I2C, SPI

RF & Simulation Tools: Ansys HFSS, PyAEDT, LTspice

PCB & Hardware Tools: KiCad, Arduino IDE, JLCPCB, Tinkercad

Frameworks & Libraries: FastAPI, Django, OpenCV, ROS2 (Fundamentals), NumPy, Matplotlib

Operating Systems: Windows, Linux (Ubuntu, Kali, Arch)

Systems & Concepts: Embedded Systems, IoT, Antenna Design, RF Fundamentals, PCB Design, Data Structures & Algorithms, Object-Oriented Programming (OOP)

Achievements

- Leetcode: (Max 1450+)
- Achieved 4th position in Robo Soccer War - high-stakes autonomous robotics competition

Leadership

- **Coordinator, House of Geeks -** Spearheaded technical workshops and hackathons for a community of 600+ student developers